Common UNIX Commands and Tools that Enhance Security

These powerful UNIX commands and tools fortify digital defences. Take a look.

In our current security system development environment, we have different tools and commands that are readily available for use by an engineer, especially the security engineer. Let's explore the most popular commands and tools among these that help us protect against or monitor potential security attacks.

Commands

There are hundreds of commands that are useful for security analysts. Here are a few most commonly used ones.

- *ipconfig / ifconfig*
  - Shows the configuration of the assigned network interface to the server.
  - Includes MAC address, IPv4 address, IPv6 address, default gateway, transferred and received packet sizes and dropouts, MTU or maximum transmission unit that states the maximum size of the packet the network interface can transfer, and so on.
  - The *ipconfig* command is used in Windows, and *ifconfig* is for Linux/UNIX and iOS.
- *ping*
  - This is the basic command to validate the reachability of the server with the given server name or IP address.
  - It provides information on the response time to determine RTT (round trip time).
  - Usually uses ICMP protocol request and reply.
- *arp*
  - Shows the local machine's ARP (address resolution protocol) cache.
  - ARP cache displays the list of MAC addresses of the interface associated with each IP address, which the local host has communicated with recently.
  - This command can also be used to investigate a suspicious spoof attack.
- *traceroute / tracert*
  - Reports the list of hops needed to be made from source to the destination host.
  - RTT is well reported.
  - *tracert* is a Windows command and *traceroute* is a Linux-based command.
  - ICMP or UDP is used in this communication.
- *route*
  - Edits or views the host's local routing table.
  - By default, all the traffic from the host is routed via the gateway router.
  - Additional entries found in this table could be suspicious.
- *getmac*
  - Returns the MAC address and the list of network protocols associated with each address for all the network cards in the host.
- *pathping / mtr*
  - Provides information about network latency and network loss at intermediate hops between a source and destination.
  - Is a combination of *traceroute* and *ping* commands.
  - Windows uses *pathping*, while Linux uses *mtr* commands.